

Positive LDO (LT3042) — Working Document

Digital Audio Player · Rev 5

Author: Alexander Loker — design decisions, component selection, analysis and verification.

Drafted by: Claude (Anthropic) — structure and prose from the author's direction and source data.

1. Part Identification

Field	Value
Manufacturer / MPN	Analog Devices — LT3042EDD#TRPBF (package to confirm)
Package	10-lead DFN 3 × 3 mm, exposed pad (pin 11)
Function	Ultralow-noise positive LDO, +3V9 → +3.3 V analog rail

2. Design Requirements

Requirement	Value
Input	+3V9 from LTC3440, 3.827 – 3.981 V
Input ripple	7.7 mV pp nominal, ±90 mV load step (min 3.74 V)
Output	3.32 V
Load	I/V, summing, OPA1622 (V+ side), DAC analog TBD
Max output	0.6 V RMS (EW300: 119 dB/V worst nozzle, 28 Ω ±15%)
Output current	≈ 27 mA quiescent; ≈ 98 mA pk (200 mA rated)
Current limit	125 mA programmed (R_ILIM)
UVLO	3.5 V rising
Power good	≈ 91% of V_OUT, to MCU GPIO

Constraints: Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1, 2	IN	+3V9, C_IN
3	EN/UV	R_EN2 from IN, R_EN1 to GND
4	PG	R_PU to +3V3, M̄CU GPIO
5	ILIM	R_ILIM to GND (Kelvin to pin 8)
6	PGFB	R_PG2 / R_PG1 midpoint
7	SET	R_SET C_SET to GND
8	GND	GND
9	OUTS	OUT at load
10	OUT	+3V3A, C_OUT, R_PG2
11	Exposed pad	GND

4. External Components

Ref	Value	Rating / package	MPN	AL-PN
R_SET	33.2 kΩ 0.1%	0402	YAGEO RT0402BRD0733K2L	#1-085
C_SET	2.2 μF (≥ 1 μF effective at 3.32 V)	Low leakage, 0402 target	Murata TBD	—
C_IN	10 μF (6.76 μF eff. at 3.9 V)	10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
C_OUT	10 μF (7.56 μF eff. at 3.3 V)	10 V X7R 0805	Murata GCM21BR71A106KE22K	#1-081
R_EN1	50 kΩ 0.1%	0603	YAGEO RT0603BRD0750KL	#1-075
R_EN2	90.9 kΩ 0.1%	0402	YAGEO RT0402BRD0790K9L	#1-084
R_PG1	100 kΩ 0.1%	0603	YAGEO RT0603BRD07100KL	#1-066
R_PG2	909 kΩ 1%	0402	YAGEO RC0402FR-07909KL	#1-083
R_ILIM	1.0 kΩ 0.1%	0402	YAGEO RT0402BRD071KL	—
R_PU	10 kΩ	0603	Generic	—

5. Equations and Results

Output voltage

$$V_{OUT} = I_{SET} R_{SET} = 100 \mu A \times 33.2 k\Omega = 3.32 V$$

Undervoltage lockout (I_{EN} term neglected)

$$V_{IN(UVLO)} = 1.24 \left(1 + \frac{R_{EN2}}{R_{EN1}} \right) + I_{EN} R_{EN2} = 1.24 \left(1 + \frac{90.9}{50} \right) = 3.494 V$$

Power good threshold (I_{PGFB} term neglected)

$$V_{OUT(PG)} = 0.3 \left(1 + \frac{R_{PG2}}{R_{PG1}} \right) + I_{PGFB} R_{PG2} = 0.3 \left(1 + \frac{909}{100} \right) = 3.027 V \text{ (91.2\%)}$$

Current limit (ILIM threshold 300 mV; 125 mA·kΩ scale factor)

$$I_{LIM} = \frac{125 mA \cdot k\Omega}{R_{ILIM}} = \frac{125 mA \cdot k\Omega}{1.0 k\Omega} = 125 mA \text{ (1.27} \times \text{ the 98 mA peak)}$$

SET noise filter corner (1 μF effective)

$$f_c = \frac{1}{2\pi R_{SET} C_{SET}} = \frac{1}{2\pi \times 33.2 k\Omega \times 1 \mu F} = 4.8 Hz$$

Soft-start (fast start-up to ≈ 2.7 V, then R_SET·C_SET tail; 1 μF effective)

$$\tau = R_{SET} C_{SET} = 33.2 ms, \quad t_{99} = \tau \ln \frac{0.62}{0.033} = 2.93 \tau \approx 97 ms / \mu F$$

C_SET sweep (simulated, LTC3440 replaced by ideal 3.9 V source, 29 Ω load)

C_SET (effective)	t_99	Noise, 20 Hz – 20 kHz
0.47 μ F	46 ms	1.16 μ V RMS
1 μ F	98 ms	0.63 μ V RMS
2.2 μ F	214 ms	0.40 μ V RMS
4.7 μ F	455 ms	0.34 μ V RMS

Headroom

$$V_{IN(MIN)} - V_{OUT} = 3.74 - 3.32 = 0.42 \text{ V}$$

Maximum SPL (0.6 V RMS, EW300 copper nozzle 119 dB/V)

$$SPL_{MAX} = 119 + 20 \log_{10}(0.6) = 114.6 \text{ dB} \quad (116.6 \text{ dB steel nozzle})$$

Load current (0.6 V RMS = 0.85 V pk design maximum, both channels in phase)

Case	Per channel pk	Signal total	+ Quiescent	LT3042 pk
Full scale, 23.8 Ω (28 Ω –15%)	35.7 mA	71.3 mA	27 mA	$\approx$ 98 mA
Full scale, 28 Ω nominal	30.3 mA	60.6 mA	27 mA	$\approx$ 88 mA
110 dB SPL peak (0.35 V RMS), 23.8 Ω	21.1 mA	42.2 mA	27 mA	$\approx$ 69 mA

6. Important Values

Item	Value
SET pin current	100 μ A
EN/UV rising threshold	1.24 V
PGFB threshold	300 mV
Dropout	350 mV typ at 200 mA
Rated output current	200 mA
Current limit	125 mA programmed; ILIM also monitors output current (1:400, 0 – 300 mV)
Peak load	$\approx$ 98 mA at 0.6 V RMS max (49% of rating)
Minimum headroom	0.42 V
UVLO divider current	$\approx$ 28 μ A from +3V9
PG divider current	$\approx$ 3.3 μ A from OUT
Fast start-up knee	$\approx$ 2.7 V (sim, independent of C_SET)
PG high	$\approx$ 25 ms after enable (1 μ F effective)
Output settled (99%)	$\approx$ 98 ms after enable (1 μ F effective)
Output noise	0.63 μ V RMS, 20 Hz – 20 kHz (sim, 1 μ F effective)
PG in shutdown	High-Z — reads good while EN/UV low (sim)

7. Open Problems

#	Problem
1	Max dropout at peak load not verified; ILIM scale-factor tolerance — confirm minimum limit stays above 98 mA; confirm R_ILIM MPN
2	EN/UV threshold tolerance, hysteresis and I_EN not included
3	I_PGFB error term not included
4	Quiescent (27 mA) unconfirmed — I/V estimate, OPA1622 I_Q at ± 3.3 V unread
5	DAC analog supply allocation to this rail open
6	C_SET MPN — confirm ≥ 1 μ F effective at 3.32 V bias (SimSurfing)
7	Confirm package (DFN vs MSOP-10E)
8	Sim with LTC3440: 27 $\leftrightarrow$ 98 mA load step, 1.2 MHz ripple at OUT (FFT)
9	Firmware: PG invalid before +3V9 is up; wait ≈ 75 ms after PG before OPA1622 enable
10	Verify simulated noise against datasheet noise vs C_SET curve

8. Revision Log

Rev	Date	Change
0	2026-09-21	Initial issue: R_SET, UVLO, PG, soft-start, load current, open problems.
1	2026-09-22	C_SET 2.2 μ F nominal (≥ 1 μ F effective); startup and noise sweeps added.
2	2026-09-22	C_IN / C_OUT to 10 μ F GCM21BR71A106KE22K; derating item closed.
3	2026-09-22	Library parts assigned: #1-081, #1-083, #1-084, #1-085. R_PG2 to 1%.
4	2026-09-22	Max output set to 0.6 V RMS (EW300); peak load 205 $\rightarrow$ 98 mA. Closed: 200 mA rating exceeded.
5	2026-09-22	Current limit programmed to 125 mA: R_ILIM 1.0 k Ω .